Phase 1: Structured Diagnosis

This document diagnoses bias in **both required cases** from the assignment brief: **Case A, the Our World in Data COVID-19 weekly cases map**, and **Case B, the Gapminder bubble chart**. The analysis is grounded in direct inspection of the interfaces and in the uploaded lecture materials. For each bias, I identify what the bias is, where it appears, what misleading impression it creates, how severe it is, and what root cause most plausibly produces it. [1] [2] [3] [4] [5] [6]

Case A: Our World in Data COVID-19 Weekly Cases Map

The standalone OWID map is visually clear and authoritative, but its defaults encourage viewers to compare countries through a **raw-total** lens instead of a more epidemiologically meaningful burden lens. The screenshot below highlights the main interface cues that support that interpretation.

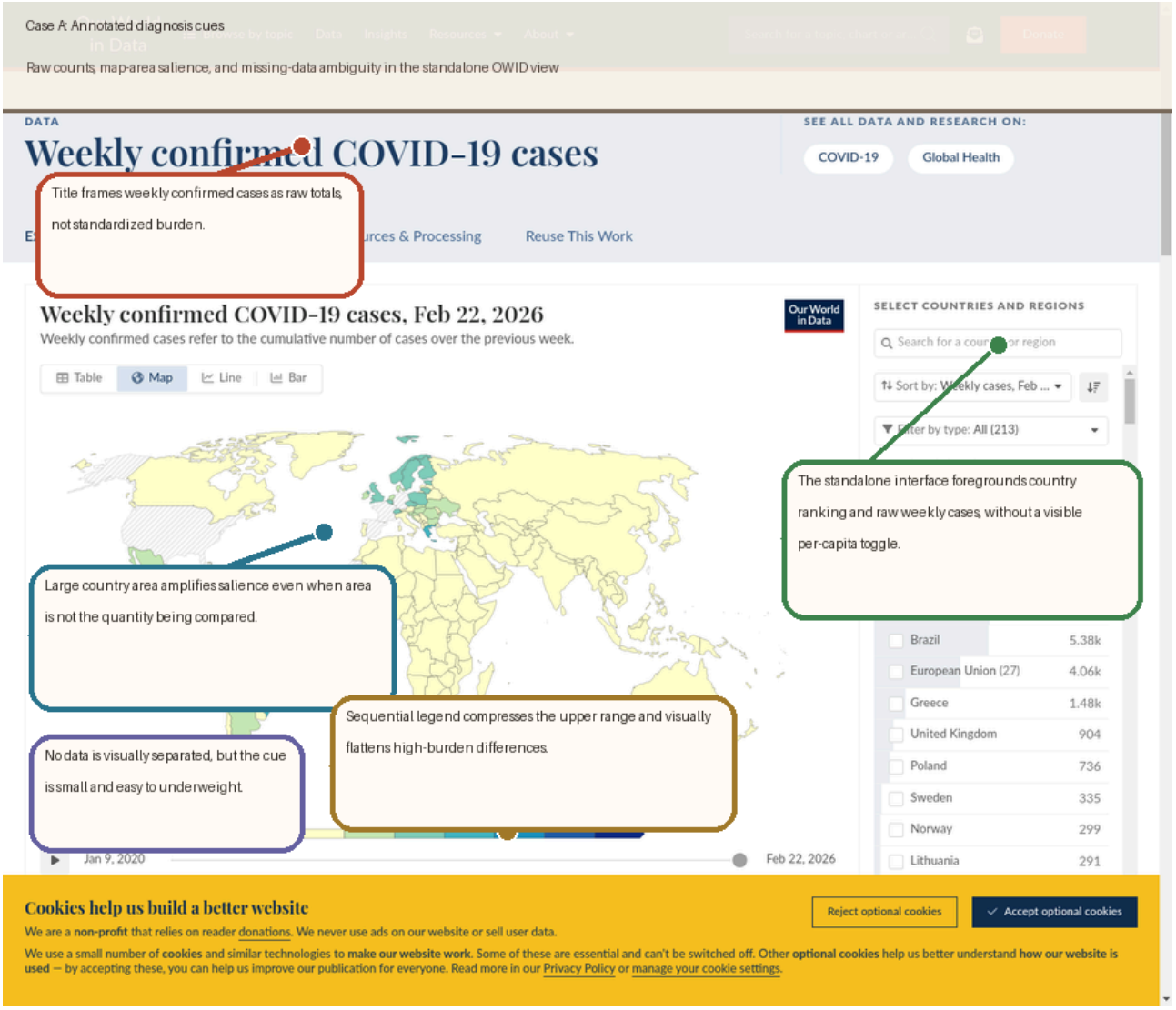

Figure 1. Annotated Case A screenshot showing the raw-count title framing, the choropleth map, the compressed sequential legend, the small no-data cue, and the country ranking panel.

Bias	Where it appears and misleading impression	Course-theory grounding	Severity	Root cause
Absolute weekly counts are framed as the default basis for comparison	The title and legend foreground **weekly confirmed COVID-19 cases** in raw counts, and the standalone view does not visibly foreground a relative-to-population alternative. This can make viewers believe that countries with larger totals are necessarily experiencing the more serious outbreak.	**L4 [Create or convert]**; **L2 [Principle of Consistency]**; **L3 [Deceptive Visualization]**	**Dangerous.** In a public-health context, a reasonable decision-maker could underweight a smaller country with a severe per-capita outbreak because the default presentation makes raw totals appear to be the correct comparison basis.	**Framing** + **interaction design**
Country area becomes a visual proxy for importance	The choropleth gives geographically large countries far more screen area than small countries. This can make large territories feel more important even though land area is not the variable under analysis.	**L2 [Choosing Visual Encodings]**; **L2 [Effectiveness Rankings]**	**Misleading.** A casual viewer can over-attend to large countries simply because they occupy more visual space, even if the intended comparison concerns disease burden rather than territorial footprint.	**Visual encoding**
Upper-end legend compression weakens	The sequential color scale culminates in a dark upper band up	**L3 [Poor Scaling]**; **L2 [Information in**	**Misleading.** The chart still identifies which countries are high,	**Visual encoding**

Bias	Where it appears and misleading impression	Course-theory grounding	Severity	Root cause
comparison among high values	to about **1 million**, which visually flattens differences among high-burden countries. This can leave the impression that all dark countries are roughly equivalent.	**Hue and Lightness]**	but it weakens discrimination where comparison is often most important.	
Missingness and measurement comparability are visually under-signaled	The legend includes a **No data** swatch, but it is visually minor, and the map does not strongly encode uncertainty arising from uneven testing and reporting practices. This can make non-reporting or weak comparability feel like low severity.	**L4 [Data Examination]; L5 [Critique your prototype]**	**Misleading.** The distortion may not directly drive a policy decision by itself, but it can give viewers a stronger sense of certainty and comparability than the data justify.	**Data** + **visual encoding**

The most severe issue in Case A is the **raw-count default**, because it shapes the first and strongest interpretation before the viewer has a chance to question whether the comparison basis is appropriate. The remaining issues reinforce that framing by making the map look more spatially and statistically straightforward than the underlying measurement conditions really are.

Case B: Gapminder Bubble Chart

Gapminder is compelling because it makes long-term development data memorable, but its default presentation is also rhetorically strong. It encourages viewers to watch countries move, infer smooth historical progress from that motion, and accept the

logarithmic income axis as a natural analytical choice rather than a consequential design decision. [1] [5] [6]

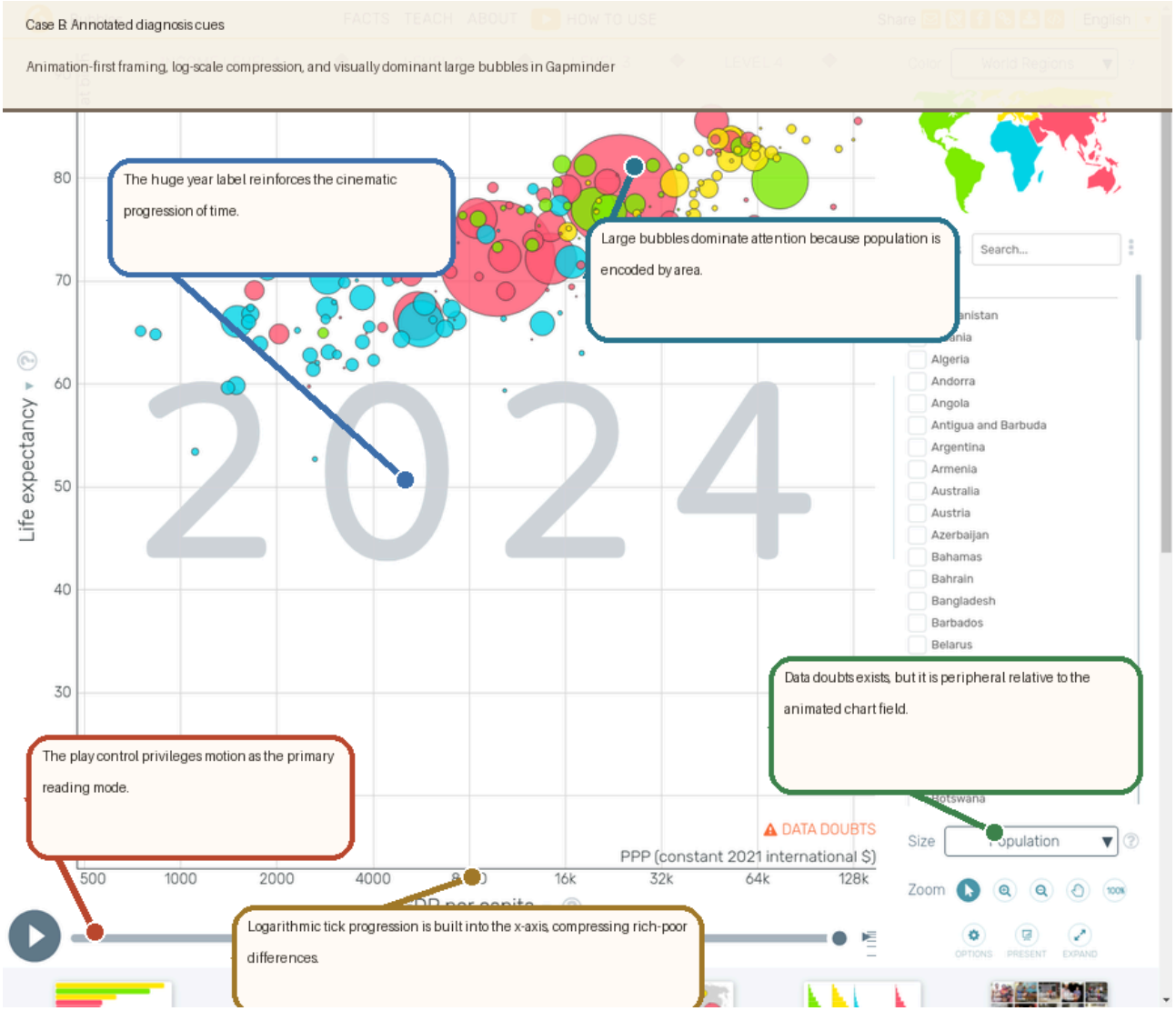

Figure 2. Annotated Case B screenshot showing the play control, oversized year staging, logarithmic income axis, visually dominant large bubbles, and peripheral uncertainty cue.

Bias	Where it appears and misleading impression	Course-theory grounding	Severity	Root cause
Animation encourages a smooth progress narrative	The play control and the large year overlay make temporal motion the primary reading mode. As bubbles drift upward and rightward, viewers are encouraged to infer coherent developmental progress and sometimes even implied causality.	**L10 [Constructing Narratives]**; **L10 [Attribution of Causality]**; **L10 [Problems of Animation]**	**Misleading.** The chart does not falsify the data, but it strongly nudges viewers toward a persuasive story that is more coherent than the evidence alone warrants.	**Interaction design** + **framing**
The interface exceeds human multiple-object tracking limits	During playback, many bubbles move simultaneously while also varying in size and color. Most viewers cannot accurately track more than a small subset of them, so the display pushes them toward impressionistic reading.	**L10 [Multiple Object Tracking]**; **L10 [When Does Animation Help?]**	**Misleading.** The interface feels analytically precise, but much of that apparent precision is lost because the viewer cannot actually follow that many moving marks at once.	**Interaction design**
The default log-scale x-axis compresses rich-poor differences	The income axis uses logarithmic spacing, visible in the tick progression from **500** to **128k**. This compresses high-end income differences and can make countries appear	**L3 [Poor Scaling]**; **L2 [Mapping Data to Visual Variables]**; **L4 [Angle, Framing and Focus]**	**Misleading.** A log scale can be analytically defensible, but when it is hidden inside the default view it privileges one interpretation while obscuring	**Visual encoding** + **framing**

Bias	Where it appears and misleading impression	Course-theory grounding	Severity	Root cause
	more convergent than they would on a linear scale.		the rhetorical effect of the choice.	
Smooth motion overstates continuity and certainty in the historical record	Playback and trails create fluid paths even though the teaching dataset is sampled discretely and not all years carry equal certainty. This can make historical development look continuously observed rather than sampled and partially regularized.	**L10 [Congruence Principle]**; **L10 [Animation vs. Small Multiples]**; **L4 [Data Examination]**	**Misleading.** The smoothness comes partly from the interface, not only from the data, so the display can overstate continuity in the evidence.	**Data** + **interaction design**

The deepest issue in Case B is therefore **compound rather than isolated**. Animation supplies narrative momentum, the log scale changes the geometry of comparison, and the interface regularizes historical change into smooth motion. Together these design choices make the visualization memorable and persuasive, but not neutral.

Diagnostic conclusion

Across the two cases, the most severe bias in **Case A** is the choice to foreground **absolute weekly case counts** in a public-health comparison, while the most important bias in **Case B** is the combination of **animation-first storytelling** and **hidden scale rhetoric**. These findings justify choosing **Case B** for redesign, because an interactive alternative can directly repair at least two diagnosed problems by replacing autoplay with comparison-first views and by exposing the effect of scale choice instead of hiding it. [1] [5] [6]

References

1. **Assignment 2 brief.** Uploaded file: `Assignment2(1).pdf`.
2. **Our World in Data – Weekly confirmed COVID-19 cases.** https://ourworldindata.org/grapher/weekly-covid-cases
3. **CS5346 Lecture 2.** Uploaded file: `CS5346-2026_Lecture2.pdf`.
4. **CS5346 Lecture 5.** Uploaded file: `CS5346-2026_Lecture5.pdf`.
5. **Gapminder Tools.** https://www.gapminder.org/tools/
6. **CS5346 Lecture 10.** Uploaded file: `CS5346-2026_Lecture10.pdf`.